GRAPHIC CARDS: HOW THEY WORK

EVER WONDERED WHAT GOES ON BEHIND THE SCENES WHEN YOU POWER UP YOUR GAME? READ ON...

Evolution of computing over the last decade has seen a distinct differentiation based on the nature of the workload. On the one hand you have the computing heavy tasks while on the other, there are graphics intensive tasks. Rewind back a decade, and you will remember almost every other motherboard having a dedicated display port as the graphics processor was onboard. Also some present day processors which sport integrated graphics also have IGP ports on the board. But with the graphical workloads getting more advanced and outputs more realistic, you would need to invest in a dedicated graphics processing unit to divide tasks between the CPU and the GPU. Another advantage of having a dedicated graphics processing unit is the fact that it lets your CPU be free to perform other tasks. For instance you can rip a movie in the background while playing a game.

We're used to seeing pixel crunching monsters that can spit out millions of polygons per second, and run even the most demanding 3D games at blistering frame rates. Yet very few of you know exactly how these marvels work. Today's graphic cards are marvels of engineering, with over 50 times the raw computational power of current CPUs, but they have far more specific and streamlined a task. There are two aspects to understanding how a graphics card works. The first is to understand how it works with the rest of the components in a personal computer to generate an image. The second part would be to understand the role of major components on a video card.